Exam 1

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Exercise 1

Convert the following linear differential equation into a system of first-order ODEs and solve for the general solution based on the identified system matrix:

$$y^{(4)} - 3y^{(2)} = -2y$$

Note: (n) refers to the n'th derivative

Given the general solution, verify (by argument and/or calculation) that this indeed forms a basis for all solutions.

linerization:

$y_1 = y$

$y_2 = y' = y^{(1)}$

$y_3 = y'' = y^{(2)}$

$y_4 = y''' = y^{(3)}$

$y^{(4)} = -2y + 3y^{(2)}$

$$\bar{\bar{A}} = \begin{vmatrix} y_1' \\ y_2' \\ y_3' \\ y_4' \end{vmatrix} = \begin{array}{c} \begin{array}{cccc} y & y^{(1)} & y^{(2)} & y^{(3)} \end{array} \\ \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 0 & 3 & 0 \end{vmatrix} \end{array} \begin{vmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{vmatrix}$$

Eigen values:

$$\det(\bar{\bar{A}} - \lambda \bar{\bar{I}}) = 0 \qquad \begin{vmatrix} 0-\lambda & 1 & 0 & 0 \\ 0 & 0-\lambda & 1 & 0 \\ 0 & 0 & 0-\lambda & 1 \\ -2 & 0 & 3 & 0-\lambda \end{vmatrix} = \lambda^4 - 3\lambda^2 + 2 \implies \lambda = \{\sqrt{2}, -\sqrt{2}, 1, -1\}$$

Eigenvectors

$$\sqrt{2} : \bar{v}_1 = (\bar{\bar{A}} - \lambda \bar{\bar{I}}) \cdot \bar{x} = \bar{0} = \begin{vmatrix} 0-\sqrt{2} & 1 & 0 & 0 \\ 0 & 0-\sqrt{2} & 1 & 0 \\ 0 & 0 & 0-\sqrt{2} & 1 \\ -2 & 0 & 3 & 0-\sqrt{2} \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \end{vmatrix} \implies \begin{vmatrix} \frac{\sqrt{2}}{4} \\ \frac{1}{2} \\ \frac{\sqrt{2}}{2} \\ 1 \end{vmatrix}$$

$$-\sqrt{2} : \bar{v}_2 = (\bar{\bar{A}} - \lambda \bar{\bar{I}}) \cdot \bar{x} = \bar{0} = \begin{vmatrix} 0+\sqrt{2} & 1 & 0 & 0 \\ 0 & 0+\sqrt{2} & 1 & 0 \\ 0 & 0 & 0+\sqrt{2} & 1 \\ -2 & 0 & 3 & 0+\sqrt{2} \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \end{vmatrix} \implies \begin{vmatrix} -\frac{\sqrt{2}}{4} \\ \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \\ 1 \end{vmatrix}$$

$$\begin{vmatrix} -2 & 0 & 3 & 0+\sqrt{2} \end{vmatrix} \quad \begin{vmatrix} x_4 \end{vmatrix} \quad \begin{vmatrix} 0 \end{vmatrix} \qquad \begin{vmatrix} -\frac{\sqrt{2}}{2} \\ 1 \end{vmatrix}$$

$$1 \ : \bar{v}_3 = \left(\bar{\bar{A}} - \lambda \bar{\bar{I}}\right) \cdot \bar{x} = \bar{0} \quad = \begin{vmatrix} 0-1 & 1 & 0 & 0 \\ 0 & 0-1 & 1 & 0 \\ 0 & 0 & 0-1 & 1 \\ -2 & 0 & 3 & 0-1 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 \\ 1 \\ 1 \\ 1 \end{vmatrix}$$

$$-1 \ : \bar{v}_4 = \left(\bar{\bar{A}} - \lambda \bar{\bar{I}}\right) \cdot \bar{x} = \bar{0} \quad = \begin{vmatrix} 0+1 & 1 & 0 & 0 \\ 0 & 0+1 & 1 & 0 \\ 0 & 0 & 0+1 & 1 \\ -2 & 0 & 3 & 0+1 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \\ 0 \end{vmatrix} \Rightarrow \begin{vmatrix} -1 \\ 1 \\ -1 \\ 1 \end{vmatrix}$$

General Solution:

$$y(t) = C_1 \bar{v}_1 e^{\lambda t} + C_2 \bar{v}_2 e^{\lambda t} + C_3 \bar{v}_3 e^{\lambda t} + C_4 \bar{v}_4 e^{\lambda t}$$

$$= C_1 \begin{vmatrix} \frac{\sqrt{2}}{4} \\ \frac{1}{2} \\ \frac{\sqrt{2}}{2} \\ 1 \end{vmatrix} e^{\sqrt{2}t} + C_2 \begin{vmatrix} -\frac{\sqrt{2}}{4} \\ \frac{1}{2} \\ -\frac{\sqrt{2}}{2} \\ 1 \end{vmatrix} e^{-\sqrt{2}t} + C_3 \begin{vmatrix} 1 \\ 1 \\ 1 \\ 1 \end{vmatrix} e^{1t} + C_4 \begin{vmatrix} -1 \\ 1 \\ -1 \\ 1 \end{vmatrix} e^{-1t}$$

Verify that is on the form of a basis:

Sæt eigen vektorene sammen til en eigenmatrix og så lav RREF. Hvis den ender på RREF, så ved vi at det former en basis.

$$\begin{vmatrix} \frac{\sqrt{2}}{4} & -\frac{\sqrt{2}}{4} & 1 & -1 \\ \frac{1}{2} & \frac{1}{2} & 1 & 1 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 1 & -1 \\ 1 & 1 & 1 & 1 \end{vmatrix} \overset{RREF}{\Longrightarrow} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Exercise 2

For each of the following systems of equations, use Gauss elimination to find the solution(s), in case there are any, and explain why not in case there isn't:

a)
$$\begin{aligned} x_1 + 2x_2 - x_3 &= 9 \\ x_1 + 6x_3 + 8x_2 &= -2 \\ -2x_1 - 6x_3 &= 38 \end{aligned}$$

b)
$$\begin{aligned} 6x_1 + 4x_3 &= 10 \\ x_2 + 3x_3 &= 5 \\ -3x_1 + 2x_2 + 4x_3 &= 5 \end{aligned}$$

c)
$$\begin{aligned} 4x_2 + 3x_3 &= 7 \\ -x_3/2 + x_1 &= 2 \\ 2x_2 + 3x_1 &= 3 \end{aligned}$$

In your answer, identify the coefficient matrix and the augmented matrix, and for the coefficient matrix its rank and nullity.

Note: It is sufficient that the Gauss elimination procedure is detailed for one of the systems a), b) or c)

a)

$$\underbrace{\begin{vmatrix} 1 & 2 & -1 \\ 1 & 8 & 6 \\ -2 & 0 & -6 \end{vmatrix}}_{\text{Coefficient matrix}} \qquad \underbrace{\left|\begin{array}{ccc:c} 1 & 2 & -1 & 9 \\ 1 & 8 & 6 & -2 \\ -2 & 0 & -6 & 38 \end{array}\right|}_{\text{Augmented matrix}}$$

$$\begin{vmatrix} 1 & 2 & -1 \\ 1 & 8 & 6 \\ -2 & 0 & -6 \end{vmatrix} \xrightarrow{REF} \begin{vmatrix} 1 & 2 & -1 \\ 0 & 6 & 7 \\ 0 & 0 & -38/3 \end{vmatrix} \quad \text{rank} = r = 3$$

$$\left|\begin{array}{ccc:c} 1 & 2 & -1 & 9 \\ 1 & 8 & 6 & -2 \\ -2 & 0 & -6 & 38 \end{array}\right| \xrightarrow{REF} \begin{vmatrix} 1 & 2 & -1 & 9 \\ 0 & 6 & 7 & -11 \\ 0 & 0 & -38/3 & 190/3 \end{vmatrix} \quad \text{rank} = \tilde{r} = 3$$

Følg flowchart:

$r < \tilde{r}$?

↓ nej $r = \tilde{r}$

↓

$Ax = ?$

↓ $Ax = b$

$r < n$?

↓ $r = n$

1 løsning

Find løsning

$\bar{\bar{A}}\,\bar{x} = \bar{b} = \quad 1 \quad 2 \quad -1$

1 8 6
-2 0

Exercise 3

Determine the critical points of the following differential equation:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}' = \begin{bmatrix} cos(y_2) \\ 3y_1 \end{bmatrix}$$

Note: x' refers to the derivative of x

For each of the critical points, illustrate the location in the phase plane and determine the stability and type of the critical point. Illustrate an example trajectory of the phase portrait for one of the critical points, including indication of its (tangent) direction.

Step 1: Find Critical Points

Critical points occur where the derivatives are zero:

$$\cos(y_2) = 0, \quad 3y_1 = 0 \Rightarrow y_1 = 0, \quad \cos(y_2) = 0 \Rightarrow y_2 = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \dots$$

So the critical points are:

$$(0, \pm\frac{\pi}{2}),\ (0, \pm\frac{3\pi}{2}), \dots$$

Step 2: Linearize Around Each Critical Point

Let's linearize using the Jacobian matrix $J(y_1, y_2)$:

$$J(y_1, y_2) = \begin{bmatrix} \frac{\partial}{\partial y_1}[\cos(y_2)] & \frac{\partial}{\partial y_2}[\cos(y_2)] \\ \frac{\partial}{\partial y_1}[3y_1] & \frac{\partial}{\partial y_2}[3y_1] \end{bmatrix} = \begin{bmatrix} 0 & -\sin(y_2) \\ 3 & 0 \end{bmatrix}$$

Now evaluate J at the critical point $(0, \frac{\pi}{2})$:

$$\sin\left(\frac{\pi}{2}\right) = 1 \Rightarrow J = \begin{bmatrix} 0 & -1 \\ 3 & 0 \end{bmatrix}$$

Step 3: Find Eigenvalues

Find the eigenvalues of J:

$$\det\begin{bmatrix} -\lambda & -1 \\ 3 & -\lambda \end{bmatrix} = \lambda^2 + 3 = 0 \Rightarrow \lambda = \pm i\sqrt{3}$$

Pure imaginary eigenvalues → this is a **center** (non-spiraling closed orbits), and the linearized system is **neutrally stable**.

Step 3: Find Eigenvalues

Find the eigenvalues of J:

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Step 4: Repeat for $y_2 = -\frac{\pi}{2}$

$$\sin\left(-\frac{\pi}{2}\right) = -1 \Rightarrow J = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$$

$$\det \begin{bmatrix} -\lambda & 1 \\ 3 & -\lambda \end{bmatrix} = \lambda^2 - 3 = 0 \Rightarrow \lambda = \pm\sqrt{3}$$

→ Real eigenvalues with opposite signs → **saddle point**, which is **unstable**.

Summary of Results

Critical Point	Type	Stability
(0, π/2), (0, 3π/2), ...	Center	Neutrally stable
(0, -π/2), (0, -3π/2),...	Saddle	Unstable

Step 5: Sketch Phase Portrait

- At centers (e.g., (0, π/2)) draw closed orbits around the point.
- At saddle points (e.g., (0, -π/2)) draw hyperbolic trajectories with stable and unstable manifolds.
- Use the vector field:
  - $\dot{y}_1 = \cos(y_2)$
  - $\dot{y}_2 = 3y_1$
- At $y_2 = 0$, $\cos(y_2) = 1$, so $\dot{y}_1 = 1$ – pointing right
- At $y_2 = \pi$, $\cos(y_2) = -1$ – pointing left

I can generate an accurate sketch of the phase portrait including an example trajectory if you'd like. Let me know.